

Critical Thinking Rubric
 Weight of assessment in final grade: 20%
 Type of assessment: individual
 Evaluator: Hegias

	Learning Outcome (numbers based on overall outcomes)	Criteria	Fail	Pass	Comments
Communicate	1.e. Apply interview and listening strategies.	Describe experiences with / thoughts on: - clear and open questions (e.g., with the technique of non-directive questioning) - reflective listening (based on the LISTEN approach) - adapting to the context of the interview	No or hardly any reflection on interview and listening skills is provided	Experiences with interviewing or listening skills are included and elaborated on.	
Collaborate	2.b. Explain how different backgrounds (including education and science) play a role in collaboration	- The first part of the team contract (the student's knowledge, skills, methods and values added to the team) - What did the student learn from other team members / other teams / supervisors / coaches	The critical thinking journal lacks reflection on any of the following points: - the student's knowledge, skills, methods and values added to the team, - What did the student learn from other team members / other teams / supervisors / coaches Or the reflection is superficial and only scratches the surface.	The critical thinking journal contains detailed reflections on the following points: - the student's knowledge, skills, methods and values added to the team, especially in the light of the others' knowledge, skills, methods and values. - What did the student learn from other team members / other teams / supervisors / coaches	
	2.b Reflect on transdisciplinary collaboration in such a way that it can improve collaboration.	- Challenges the team encountered - How they solved these - What did they learn from these	The critical thinking journal lacks reflection on any of the following points: - Challenges the team encountered - How they solved these - What did they learn from these Or the reflection is superficial and only scratches the surface.	The critical thinking journal contains detailed reflections on all the following points: - Challenges the team encountered - How they solved these - What did they learn from these	
	2.c. Make a theory-based standpoint on the ethical dilemmas around credit, ownership, and group work	- The way the team worked together (collaboration, cooperation or coordination) - The role(s) of the student in the team and project -	The critical thinking journal lacks reflection on any of the following points: - The way the team worked together (collaboration, cooperation or coordination) - The role(s) of the student in the team Or the reflection is superficial and only scratches the surface.	The critical thinking journal contains detailed reflections on all the following points: - The way the team worked together (collaboration, cooperation or coordination) - The role(s) of the student in the team - Giving credit -	
Research	3.b. Critically evaluate research strategies and if necessary modify and adapt these strategies	Research insight	Lacks a comprehensive understanding of the research topic. Fails to critically think about research processes and methods.	Has a comprehensive understanding of the research project and is able to think critically about methods being used.	
	3.d. Analyse outcome of an experiment in a critical way	Data analysis	Struggles with identifying problems, did not show ability to critically think about data analysis methodologies	Identifies problems, has long-term vision, and reflects on challenges in data analysis.	
	3.e apply research ethics around reproducibility, data management and	Research ethics	Struggles with identifying problems. Did not show understanding of ethics in reproducibility,	Understands problems, is able to demonstrate understanding of ethics in reproducibility, data	

	integrity		data management and integrity.	management and integrity.	
	3.f Understand processes involved with institutional review, grant applications and scientific publications	External review	Does not demonstrate understanding of scientific review processes	Understands processes of scientific review including research integrity, grants and publications	
	33.g. Apply ethical and philosophical considerations in performing research	Able to put research in ethical and philosophical context	Does not demonstrate understanding of ethical and philosophical considerations in research.	Identifies problems, has long-term vision, and reflects on Ethical and philosophical context of research.	
Reflect	4.a. Use provided frameworks to reflect on mistakes that were made in the minor	Describe what mistakes you identified within CSBB minor and what could be learned from them.	No or hardly any reflection is provided on the mistakes made in the minor.	Reflection on mistakes in the minor are provided and it has been indicated what can be learned from the mistakes.	
	4. b. Understand and apply the concept of perspective agility	Use and reflect on perspective agility. Describe the yield of perspective agility.	No or hardly any reflection is provided on the benefit of using perspective agility.	Reflection is provided on perspective agility and what yield it offered.	
	4.c. Understand how personal biases influence work.	Describe what personal and cultural preferences influences collaboration and scientific research.	Student shows no or poor understanding of biases in relation to collaboration and science.	Student can formulate and is aware of biases how that (might) influence collaboration and science.	